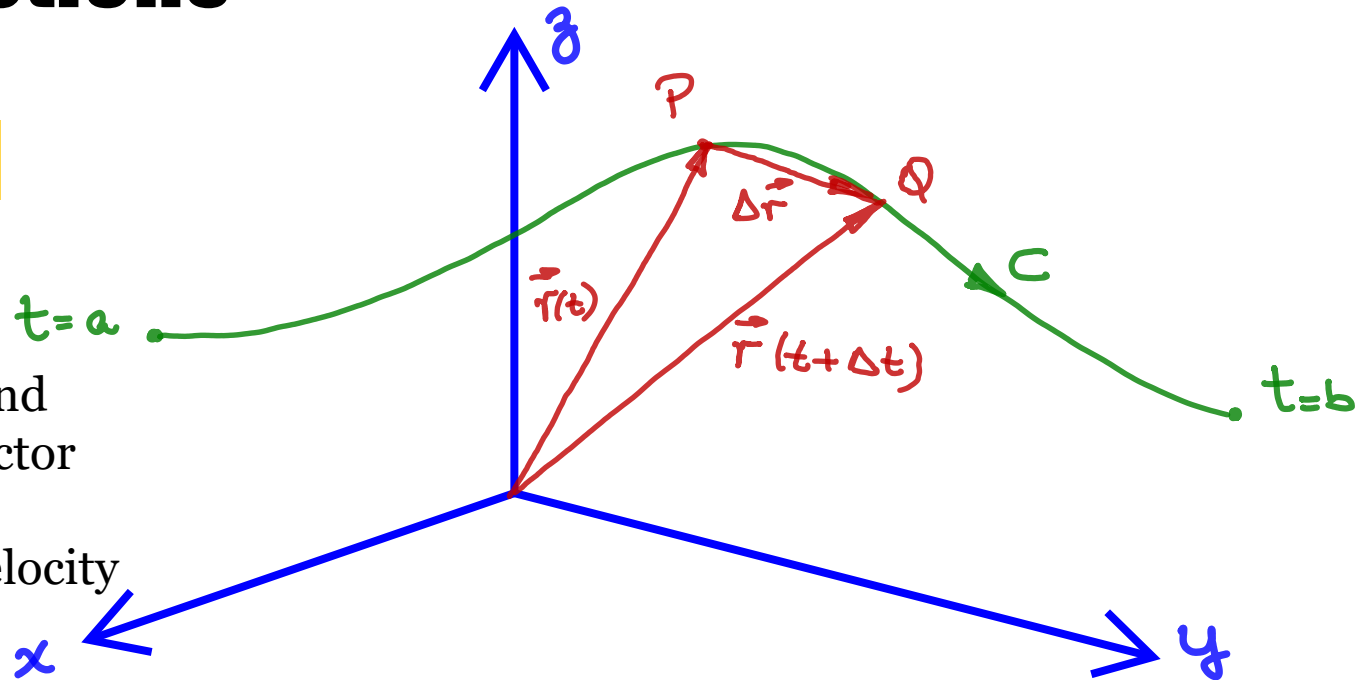


Differentiation and Integration of Vector Functions

Trim Ch 11, S. 11.9, 11.13

Outline:

1. Differentiation and Integration of vector functions
2. Displacement, velocity and acceleration



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Differentiation of Vector Functions

Let us consider the general form of a vector function:

$$\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}, \quad a \leq t \leq b$$

where $\vec{r}(t)$ is differentiable at every point of its domain. The rate of change of $\vec{r}(t)$ as a function of t is given by,

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = \frac{df(t)}{dt}\hat{i} + \frac{dg(t)}{dt}\hat{j} + \frac{dh(t)}{dt}\hat{k}$$

We can say that,

1. The vector $\frac{d\vec{r}}{dt}$, when different from $\vec{0}$, is the **vector tangent** to the curve C at P.
2. If $\frac{d\vec{r}}{dt} \neq \vec{0}$ and is continuous (i.e. $\frac{d\vec{r}}{dt}$ exists), **C** has a continuously turning tangent and **is smooth** (no sharp corners or cusps).



Problem

Find the parametric equations of the line that is tangent to the smooth curve given by,

$$\vec{r}(t) = \sin t \hat{i} + (t^2 - \cos t)\hat{j} + e^t \hat{k}$$

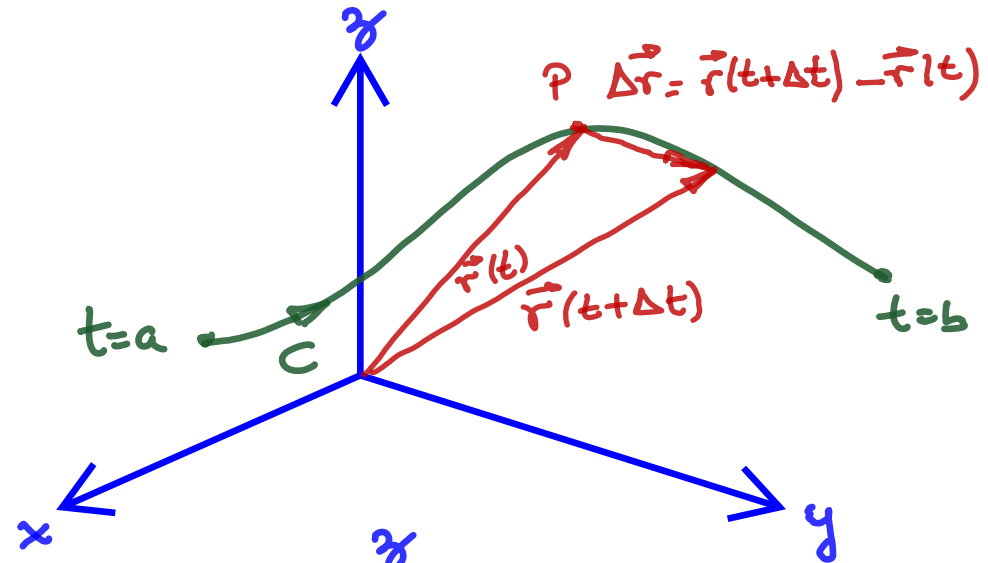
at $t = 0$.

Solution:

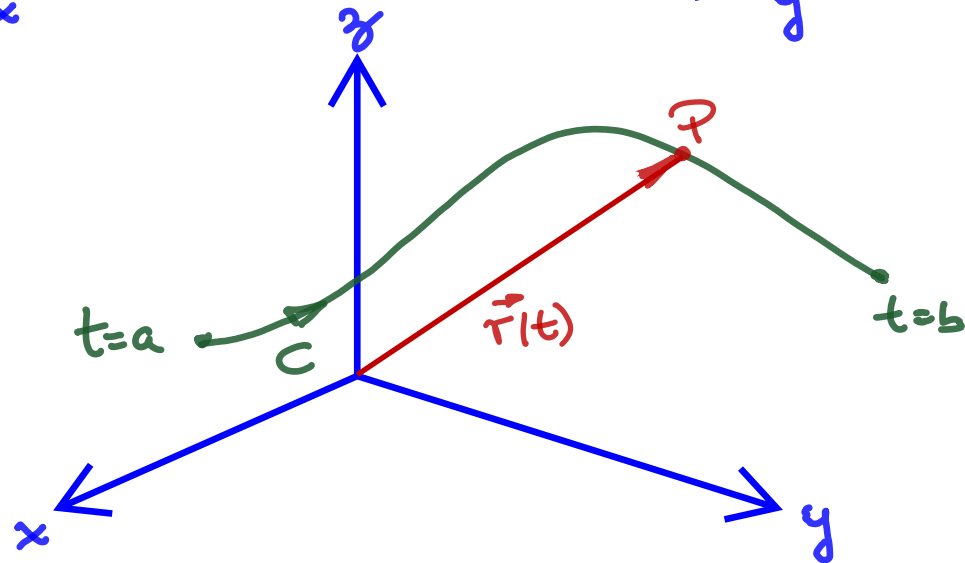
Problem

Orientation of $\frac{d\vec{r}}{dt}$

Case 1: $\Delta t > 0$



Case 2: $\Delta t < 0$



Rate of change of $\vec{r}(t)$

For a smooth curve **C** defined by the path:

$$\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}, \quad a \leq t \leq b$$

where f , g , and h are differentiable functions,

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = \frac{df(t)}{dt}\hat{i} + \frac{dg(t)}{dt}\hat{j} + \frac{dh(t)}{dt}\hat{k}$$

- $\frac{d\vec{r}(t)}{dt}$ always points in the direction in which the parameter “t” increases
- $\frac{d\vec{r}(t)}{dt}$ gives the rate of change of position versus time
- $\frac{d\vec{r}(t)}{dt}$ is never zero; and the particle does not stop or reverse direction

$\frac{d\vec{r}(t)}{dt}$ is just what we need for modeling a particle's velocity!



Velocity, Speed and Acceleration

If $\vec{r}(t)$ is the position vector of a particle moving along a smooth curve C in space; then

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt}$$

is the particle **velocity vector**, tangent to the curve at any time t and the direction of $\vec{v}(t)$ defines the direction of motion.

Speed is the magnitude of the velocity:

$$|\vec{v}(t)| = \sqrt{\left(\frac{df}{dt}\right)^2 + \left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2}$$

Acceleration is the rate of change of the velocity with respect to time:

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = \frac{d^2\vec{r}(t)}{dt^2}$$

Rules of Differentiation of Vector Functions

Theorem 11.9: Trim Chapter 11, Page 758

If $f(t)$ is a differentiable scalar function of t , and $\vec{u}(t)$ and $\vec{v}(t)$ are differentiable vector-valued functions of t , then,

$$\frac{d}{dt}(f\vec{v}) = \frac{df}{dt}(\vec{v}) + f \frac{d\vec{v}}{dt}$$

$$\frac{d}{dt}(\vec{u} \cdot \vec{v}) = \vec{u} \cdot \frac{d\vec{v}}{dt} + \frac{d\vec{u}}{dt} \cdot \vec{v}$$

$$\frac{d}{dt}(\vec{u} \times \vec{v}) = \vec{u} \times \frac{d\vec{v}}{dt} + \frac{d\vec{u}}{dt} \times \vec{v}$$



Integration of Vector Functions

A vector function $\vec{R}(t)$ is said to be an antiderivative of $\vec{r}(t)$, where $a \leq t \leq b$, if

$$\frac{d\vec{R}(t)}{dt} = \vec{r}(t)$$

The definite integral of $\vec{r}(t)$ is then,

$$\int_a^b \vec{r}(t) dt = \vec{R}(t) + \vec{C}$$

Where $\vec{C}$ is a vector with constant components

Problem

A projectile is fired from the origin over horizontal ground at an initial speed of $500 \frac{m}{s}$ and a launch angle of 60° .

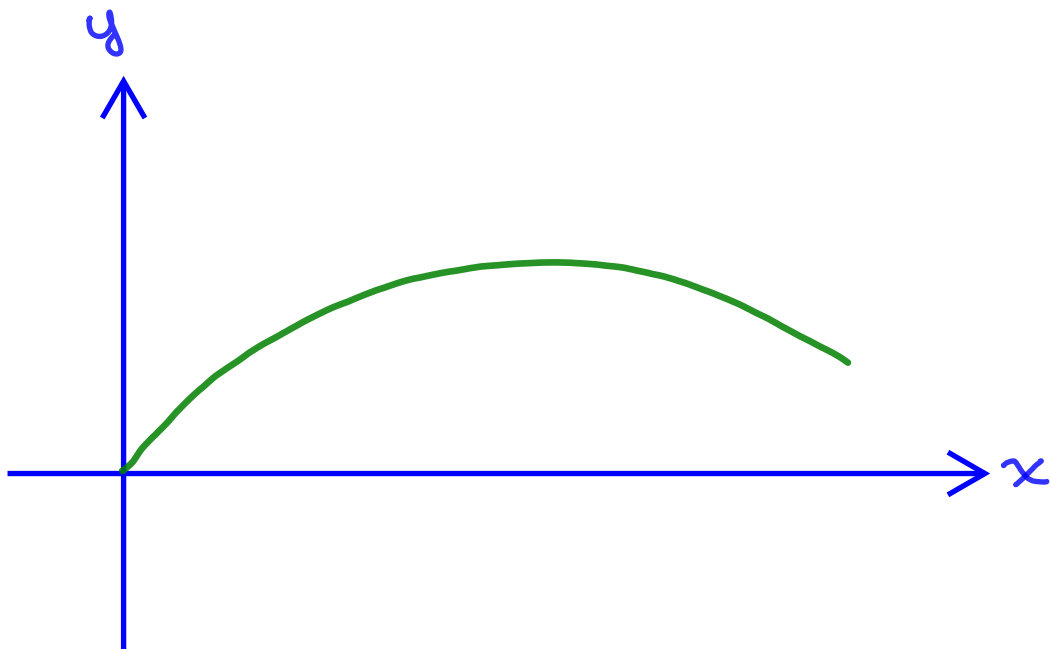
- a. Where will the projectile be 10 seconds later?
- b. Find the maximum height the projectile reaches*
- c. When does the projectile land?*

Answers:

- a. The projectile will be at $\vec{r}(t) \approx 2500\hat{i} + 3840\hat{j}$
- b. Maximum height $\approx 9566 \text{ m}$ *
- c. After $\approx 88.4 \text{ sec}$ *

* Left for your own practice at home

Solution:



Summary

- We learned how to differentiate and integrate vector functions
- We also discovered that $\frac{d\vec{r}(t)}{dt}$ is just what we need for modeling a particle's velocity!
 - Points in the direction of motion
 - Gives the rate of change of position vs. time
 - It is never zero (particle never stops or reverse direction)
- The rules of differentiation involving vectors, can also be found in Schaum's, Page 122 (20.23 – 20.27)



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Thank you